COMPANION TO SENSORS.MD · TIER 0

SENSORS_MINIMUM

A jugular compression with a blood-pressure cuff and seven electrodes — and no burr hole.

● 1 CUFF ● 4 NECK ELECTRODES ● 3 ECG ELECTRODES ● 0 INTRACRANIAL DEVICES

> **DESIGN, NOT RESULT**
>
> Nothing here was measured. **There is no impedance channel in this dataset**, so no estimator in this document has ever been run against a real electrode signal.
> Everything marked **computed** comes from `model/electrode_fim.jl`, a Fisher analysis of the model's *own* synthetic observable at the operating point the API publishes today — it says what the model thinks an electrode would identify, which is a statement about the model. Everything marked **served** is read out of `api/stream/`. Everything else is engineering argument.
> Read this as the specification for an experiment, not as its outcome.

1 THE QUESTION SENSORS.MD STOPS AT

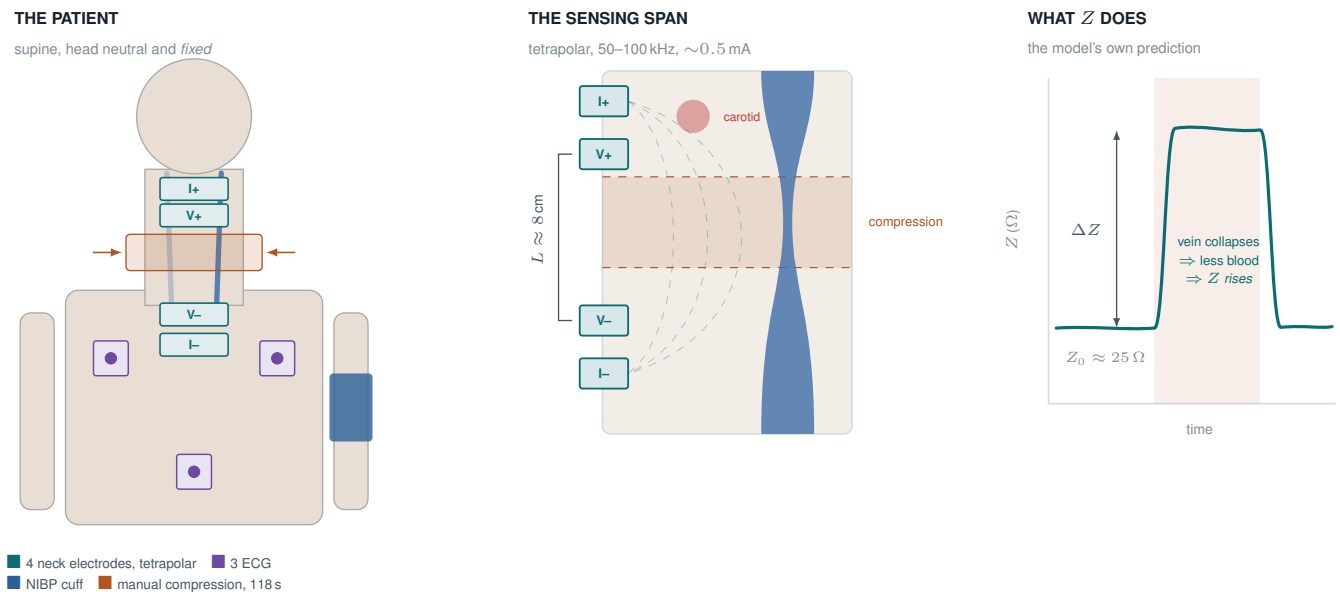
`SENSORS.md` works out that the estimator behind `api/stream/` consumes one invasive signal, one scalar and a stopwatch — and then hits a wall: intracranial pressure cannot be measured from outside the skull, so Tier 1 needs a burr hole and the neurosurgical capability to make one.

This document refuses the burr hole and keeps the manoeuvre. **What can a blood-pressure cuff and a handful of adhesive electrodes estimate about a jugular compression?**

> **THE ANSWER, IN ONE SENTENCE**
>
> The cuff and the electrodes **cannot replace the bolt** — five of the six streamed parameters go dark — but they measure, and calibrated they measure $26\times$ better than the bolt does, the one parameter the RULEBOOK says the bolt-based fit is **lying about**.

It is not a diagnostic tier. It is an **audit** tier: seven electrodes and a cuff, pointed not at the patient but at the model.

2 THE KIT



Schematic. Electrode positions indicate the anatomical region, not a placement protocol. The sense span must straddle **the compressed segment** — see rule E3.

Item	What it is	What it feeds in this repository
CUFF Blood pressure	Any NIBP monitor; a manual sphygmomanometer serves	One scalar, `abp_base`, the anchor for arterial resistance: `Ra0 = (abp_base - 25.0)/12.5` (`model/ude_cv.jl:128`). Tier 1 already needed only this number.
NECK 4 impedance electrodes	Tetrapolar array, outer pair injects, inner pair senses	**The new observable.** Neck venous volume, up to an unknown gain — the model's state `u[11]`.
ECG 3 ECG electrodes	Any 3-lead cable	Beat gating, so the impedance pulse can be ensemble-averaged. Replaces the ABP waveform's role in Tier 2.
FREE Stopwatch	—	`t_on` / `t_off`. Unchanged from `SENSORS.md`.

3 WHY THE MODEL ALREADY CONTAINS WHAT THE ELECTRODES SEE

The pleasant surprise is that no new physics is required. The compact ODE carries thirteen states, and the eleventh is the jugular cross-section:

```
u = (V_a, C_a, q_avg, Pic, Pv, V_v, Psas, P_vv, P_px, P_sys, A, P_res, P_buf)
                                                    ^^^ u[11]
A_alg = P.A0 * tube_area((j - Pxt) / P.ktube) # the tube law
dA = (A_alg - A) / P.tauA # relaxes with tau_A = 20 ms
```

A tetrapolar neck measurement is a volume measurement. Blood is the most conductive tissue in the neck, so filling the vein lowers Z and collapsing it raises Z; Nyboer's relation converts one to the other, $\Delta V = \rho\,(L/Z_0)^2\,\Delta Z$. The model's neck venous volume is simply

$$V_{\text{neck}} \;=\; A \cdot \texttt{ltube} \cdot \texttt{Ntube} \;=\; A \times 20\,\text{cm} \times 2 \text{ veins},$$

which **is** the electrode observable, up to a gain. So the question "what would electrodes identify?" can be asked of the existing model directly — and that is what `model/electrode_fim.jl` does.

> **IS THE SIGNAL EVEN THERE?** (COMPUTED)
>
> At patient 081's published operating point the model swings the neck venous volume by **158.9 mL** across the manoeuvre. Through Nyboer with $\rho = 150\,\Omega\,\text{cm}$, $Z_0 = 25\,\Omega$ and an $L = 8\,\text{cm}$ sense span, that is
>
> $$\Delta Z \;\approx\; \mathbf{4.1\,\Omega} \text{ on a } 25\,\Omega \text{ baseline.}$$
>
> Impedance cardiography routinely resolves $0.1\,\Omega$. Even if the real resting vein is a third of the model's inflated one — see §6, it almost certainly is — the swing is still $\sim 1.2\,\Omega$, an order of magnitude above a commodity noise floor. **Signal availability is not the constraint here.** Identifiability is.

4 THE GAIN PROBLEM, AND WHY IT IS ONLY HALF FATAL

An impedance electrode returns ohms, not millilitres. The conversion depends on neck geometry, electrode spacing, haematocrit and body composition — unknown per patient, and not stable enough between patients to hard-code. Two honest responses, and the design uses both:

> **(A) PROFILE THE GAIN OUT, DO NOT ASSUME IT**
>
> The observable scales linearly with the gain g, so $\partial\,\text{obs}/\partial \log g = \text{obs}$. Append that as a seventh column of the sensitivity matrix, invert the 7×7 Fisher matrix and read the six parameter variances off its diagonal. The cost of not knowing g then appears *in the numbers* instead of being wished away. The **electrode** column of the table in §5 is computed this way; **elec+calib** is the same calculation with that column removed — what a perfectly calibrated electrode would buy.

> **(B) USE A STATISTIC THE GAIN CANNOT TOUCH**
>
> The **ratio** $A_{\text{plateau}}/A_{\text{rest}}$ is gain-free: g cancels exactly. It needs no calibration, no ρ, no assumed L — only a stable Z_0 (rule E4). It is also, by luck, the quantity that decides the RULEBOOK's signature failure.

5 WHAT IS IDENTIFIABLE — COMPUTED

`model/electrode_fim.jl` builds the same Fisher matrix `stream_refit.jl:sigmas()` builds, with the same log-parameter step ($\delta = 0.05$) and the same measurement noise on ICP ($\sigma = 0.35\,\text{mmHg}$), but against three observables. Impedance noise is set at 2% of the manoeuvre's own excursion, so the electrode gets a generous fixed 50:1 SNR.

Fitted	Streamed as	ICP only needs a bolt	electrode gain unknown	elec+calib gain known	both
`E`	craniospinal compliance	0.040	78.0	75.5	0.040
`Psinus`	sagittal sinus pressure	0.004	63.0	63.0	0.004
`Ro_tot`	CSF outflow resistance	0.016	91.1	71.2	0.016
`Rvvs`	collateral outflow resist.	0.351	63.4	45.3	0.268
`k_ijvs`	**jugular resting area**	**0.077**	**0.191**	**0.003**	**0.053**
`Ca`	arterial compliance	0.034	7.18	6.97	0.034

Fractional (log-space) standard deviation. **Lower is better**; the contract calls a parameter identifiable below **0.5** (`IDENT_MAX_SIGMA`). Patient 081, final window, operating point read from `api/stream/081_batch` — the expanding-window refit.

HOW TO READ IT

- **The electrodes identify exactly one parameter of six.** `k_ijvs` at $\sigma = 0.191$; everything else lands between 7 and 91, which is two to three orders of magnitude outside the threshold. No plausible improvement in electrode noise closes a gap that size — the conclusion is not sensitive to the 2% SNR assumption.
- **Uncalibrated, the electrode matches the bolt on that parameter** (0.191 vs 0.077, same order). **Calibrated, it beats the bolt by a factor of 26** (0.003 vs 0.077). The gain is worth roughly 60× on this coordinate and nothing at all on the others.
- **The collateral is not rescued.** This was the hypothesis going in — that electrodes would see the flow diverted into the vertebral plexus — and the computation says no ($\sigma = 63$ alone, $0.351 \rightarrow 0.268$ jointly). The reason is mechanical: at the fitted point the vein is already 97.8% collapsed, so $R_j \propto A^{-2}$ is effectively infinite and the neck volume is pinned at its floor no matter what the collateral does. A sensor placed *above* the compression would see the pooling instead — but the model has no state for it (rule E3).
- Re-run at the other published operating point for 081 (the filter stream, §8) the numbers move but the verdict does not: `k_ijvs` $\sigma = 0.161$, every other parameter between 16 and 713.

6 WHAT THAT ONE PARAMETER IS WORTH

`jugular_resting_area` is not an incidental sixth of the output. It carries `bound_provenance: "qc"`, and its own metadata says why:

> **API/PARAMS/JUGULAR_RESTING_AREA.JSON** (SERVED)
>
> *"STREAMED AS QUALITY CONTROL, NOT AS A DIAGNOSIS . . . the fit drives this to 5–12 cm² (about 205 mL of jugular) which is roughly ten times physiological, and that this artefact is what causes the model's ICP pulse-amplitude failure.* ***A bedside ultrasound falsifies it in thirty seconds.*** *Read a value outside the healthy band as evidence the* FIT *is wrong, not that the patient is."*

Four neck electrodes are the version of that ultrasound which needs no operator, no probe and no training — and which runs continuously, through the manoeuvre, instead of once. Here is what it would be falsifying. Every value below is real output, read from `api/stream/`:

Patient	Jugular resting area cm²/vein, final window	vs. healthy ceiling 2.0	Collateral resistance mmHg/(mL/s)	Status
081 (batch)	4.50	2.2×	14.46 PATH	R1 FIRES
081 (filter)	3.08	1.5×	2.01 UNIDENT	MASKED
083	2.55	1.3×	22.73 UNIDENT	BORDERLINE
137	4.47	2.2×	12.11 UNIDENT	PATH. AREA
150	8.32	4.2×	19.06 UNIDENT	PATH. AREA

Every patient, every estimator: the resting jugular comes out above the physiological ceiling, by between 1.3 and 4.2 times. RULEBOOK R1 explains why — the fit cannot make the vein collapse hard enough to produce the measured ICP step, so it buys the step by inflating the resting vein and blocking the collateral instead. **Neither number is a statement about the patient.**

A neck electrode array settles this. It has two shots:

- **The absolute area**, which needs the calibration and is the decisive test: 162 mL of resting jugular blood in a 20 cm segment is either there or it is not.
- **The gain-free collapse ratio**, which needs nothing. At 081's published point the model collapses the vein to **2.2% of resting** *(computed)*. That is a strong claim about a manoeuvre that is, after all, a pair of thumbs — but it is a *weaker* test than the first, because near-complete collapse under firm manual compression is not by itself implausible. Say so plainly: **the free test is the weaker one.**

7 THE IMPLEMENTATION

Six changes. Five are small, and the sixth is a decision rather than a change.

1 — THE SOLVER STORES A THIRD STATE

`solve_run` keeps `u[4]` and `u[5]` and discards the rest. The electrode observable needs `u[11]`. This is the whole physics change:

```
icp = Vector{T}(undef, nout); pv = Vector{T}(undef, nout)
ar = Vector{T}(undef, nout) # + jugular area
...
    k += 1; tout[k] = t; icp[k] = u[4]; pv[k] = u[5]; ar[k] = u[11]
```

2 — A SECOND RESIDUAL BLOCK, WITH ITS OWN NOISE SCALE

```
# neck venous volume the electrodes see, mL: both veins over the modelled segment
neckvol(a) = a .* P.ltube .* P.Ntube

function resid(θ, t_end, tiers)
    t, icp, ar, ok = solve_obs(mkP(θ), REC); ok || return nothing
    m = (REC.tm .>= -110.0) .& (REC.tm .<= t_end) # the window, unchanged
    ix = [clamp(round(Int, (τ + 120.0)*100) + 1, 1, length(icp)) for τin REC.tm[m]]
    vcat(:bolt in tiers ? (icp[ix] .- REC.icpm[m]) ./ σ_icp : Float64[],
         :elec in tiers ? (neckvol(ar[ix]) .- REC.vneck[m]) ./ σ_vol : Float64[])
end
```

Blocks are pre-divided by their own σ, so the two observables are commensurable and either can be dropped by leaving it out of `tiers`. **The electrode tier is the same estimator called with `tiers = (:elec,)`** — where `REC.vneck` is the gain-corrected impedance channel that does not yet exist.

3 — THE GAIN IS PROFILED, NOT FITTED

Do not add g to `PNAMES`. It is not a physiological quantity and it would collect every modelling error that scales with amplitude. Append its column to the sensitivity matrix inside `sigmas()` and drop it after the inversion:

```
S = hcat(S, vol0) # d obs / d log g == obs, for a linear gain
sigma = sqrt.(max.(diag(pinv((S'*S) ./ sigma_meas^2)), 0.0))[1:6]
```

4 — THE STREAMING CONTRACT DOES NOT CHANGE

This is the part worth pausing on. `model/stream_contract.jl` fixes the six ids, the derivations, the bounds and the exact bytes of a record — and *none of it needs to move*. `emit_record` already emits `"status": "unidentifiable"` with `null` confidence bounds when σ is unavailable, which is precisely what five of the six parameters must now say. The electrode tier is not a new contract; it is the existing contract, honestly filled in.

5 — THE API SURFACE DOES NOT CHANGE EITHER

Same routes, same record schema, same component translation. A client cannot distinguish an electrode-tier stream from a bolt-tier one except by reading `identifiable` — which is the flag `CLAUDE.md` already instructs you to report *instead of* the point estimate. The discipline the API was built with is exactly what makes a degraded sensor tier serveable without a schema change.

One caveat, and it is R9: an electrode estimator writing `api/stream/<patient>/` is a **second writer**. It must take `take_stream_lock`, or write to its own `_SDIR` — as `081_batch` and `081_filter` already do.

6 — ONE METADATA DECISION, WHICH IS NOT OURS TO MAKE SILENTLY

If an electrode array actually measures the jugular area, then `jugular_resting_area`'s `bound_provenance` is no longer `"qc"`. But it should **not** become `"clinical"` either: an impedance-derived area is a device-dependent surrogate with a calibration in it, not a published human range. The honest move is a fourth provenance value — `"sensor"` — and a matching sentence in `CLAUDE.md`'s "what the numbers do and do not mean". Adding a provenance value is an API contract change and belongs in `model/api_spec.jl`, which rewrites `api/index.json`.

8 AN INCIDENTAL FINDING, WHILE WE WERE IN THERE

`model/electrode_fim.jl` reads its operating point back out of the published stream. Doing that for 081 turned up something that has nothing to do with electrodes:

> **API/STREAM/081 IS BYTE-IDENTICAL TO API/STREAM/081_FILTER**
>
> All seven files, `cmp`-identical. `CLAUDE.md` describes `081_filter` as a different experiment, "not the canonical refit" — which implies the canonical slot holds the refit. It does not; it holds the filter run.
> It matters because the filter run is the one RULEBOOK R7 indicts by name: `collateral_outflow_resistance` sits at **2.6058** for the first **117** windows — the Model 5 starting value, not departed until the compression arrives at $t = -2\,\mathrm{s}$ — while `rmse` runs from 1.15 down to 0.17 and `n_solves` stays at **39 in all 418 windows**. And the two runs disagree about the finding: `081_batch` reports the collateral **pathological at 14.46 and identifiable**, which fires R1 cleanly alongside a

$4.50\,\text{cm}^2$ jugular. The served `081` reports **2.01, unidentifiable** — R1 does not fire, and the artefact is invisible. **Serving the frozen run hides the very failure the rulebook exists to expose.** Worth a look independently of anything in this document.

9 WHAT GOES DARK, AND WHY IT STAYS DARK

Five of the six go dark, and one of those does not merely become unidentifiable — it becomes *underivable*, which is a different and worse thing:

Parameter	Loss	Why
craniospinal compliance	UNDERIVABLE	Its derivation is `1/(E * (icp_base - 5.0))`. Even with a miraculous `E`, the map to the clinical quantity divides by a **measured resting ICP** the electrode tier does not have. Two failures, not one.
CSF outflow resistance	UNIDENTIFIABLE	Lives in the slow ICP relaxation. There is no venous-volume signature of it ($\sigma = 91$).
sagittal sinus pressure	UNIDENTIFIABLE	A pressure. $\sigma = 63$.
arterial compliance	DEGRADED	$\sigma = 7.2$ from the mean-volume observable alone. The impedance *pulse* — ECG-gated, ensemble-averaged against the cuff pulse — is a genuine volume-per-pressure ratio and is not in this calculation, which uses the 1 Hz mean only. Treat the 7.2 as a floor on what is knowable, not a verdict on the pulse channel.
collateral outflow resist.	UNIDENTIFIABLE	§5. Occluded veins tell you nothing about where the flow went.

There is no clever recovery available here. Resting ICP is nearly blind to the venous outflow path in one direction, and the venous volume is nearly blind to intracranial mechanics in the other. **They are complementary observables, not substitutable ones.** The right conclusion is not "electrodes instead of a bolt" but "electrodes *alongside* a bolt, where a bolt is justified — and electrodes *alone* to audit the model, where it is not."

10 AN ELECTRODE RULEBOOK

`RULEBOOK.md` lists the parameter combinations that indict the model. A sensor tier needs its own, because these failures are not visible in any parameter value.

	Trigger	What it means
E1	Head moves during the run	Jugular cross-section is posture-dependent — that is in the metadata's own `bound_source` string. A head turn changes A without changing anything the model represents. **Fix the head, and record that you did.**
E2	The plateau drifts	Respiratory venous modulation sits at 0.2–0.3 Hz and is removed by the 1 Hz mean, but a *change in breathing pattern* across a 118 s manoeuvre is not separable from a change in venous return. The bolt tier had this too; the electrode tier is worse, because the observable is a volume.
E3	Sense span not over the compression	**The model has no spatial resolution and the electrode does.** `ltube` is one lumped 20 cm collapsible tube under a uniform external pressure. A real compression is applied at one level: the vein collapses *at* it and distends *above* it. A span placed above the thumbs measures pooling — a real and interesting signal — which the model has no state for and will absorb into A with the wrong sign.
E4	Z_0 drifts	The gain-free ratio is only free if Z_0 is stable. Electrode–skin impedance moves with sweat, gel drying and movement; the ratio then moves with the vein standing still. Log Z_0 and reject the run if it drifts more than the manoeuvre's own ΔZ.
E5	The span is not purely venous	It contains carotid, muscle, skin and air. The gain is not merely unknown — it is not a pure venous gain. Profiling g absorbs a constant offset; it does not absorb an arterial component that moves with the cardiac cycle. This is what the ECG gating is for.
E6	A second writer	R9, unchanged. An electrode estimator writing an existing patient directory truncates and interleaves silently.

11 VERDICT

TIER 0 — ONE CUFF, SEVEN ELECTRODES, NO BURR HOLE

Delivers: one of six streamed parameters, `jugular_resting_area`, at accuracy comparable to the bolt uncalibrated and $26\times$ better calibrated; a continuous gain-free collapse ratio through the manoeuvre; a plausible route to the arterial-compliance pulse channel via ECG-gated impedance. **No schema change, no route change, no new physics.**
Does not deliver: intracranial pressure, or anything derived from it. Five parameters unidentifiable, one of them underivable in principle.
Is best understood as: a continuous, operator-free version of the bedside ultrasound that `api/params/jugular_resting_area.json` already names as the way to falsify this model's signature artefact — an instrument aimed at the fit rather than the patient.

Provenance. **Computed** — the identifiability table, the volume swing, the collapse ratio, the Nyboer estimate — is output of `model/electrode_fim.jl`, both published estimators for 081. It is a Fisher analysis of a *synthetic* observable derived from the model's own state `u[11]`: it reports what the model believes an electrode would identify. **Served** — §6's per-patient table, §8's byte-identity — is read from `api/stream/` and `api/params/` as they stand on disk. **Everything else is engineering argument:** no electrode estimator has been written, and the noise level, ρ, Z_0 and L are literature-typical assumptions, stated so they can be disagreed with. **No healthy controls exist in this dataset** — every bound is from literature. The underlying model overestimates ICP pulse amplitude under compression by $\sim 7\times$, the artefact this tier is aimed at. Not clinical advice.